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Belief Elicitation with a Synchronized Lottery Choice Menu That Is Invariant to Risk Attitudes†

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This paper uses a Bayesian information processing task to compare belief elicitation mechanisms including a quadratic scoring rule, a Becker-DeGroot-Marschak pricing procedure, and a two-stage menu of lottery choices that is structured to identify a precise point of probability indifference. The choice menu yields a higher incidence of correct Bayesian responses and lower belief error averages. Unlike the quadratic scoring rule, the binary payoffs for the lottery choice mechanism are synchronized to provide theoretical incentive-compatibility regardless of risk attitudes. In addition, the choice menu structure is more transparent and intuitive than the Becker-DeGroot-Marschak procedure. (*JEL* C91, D44, D81, D83)

Payoffs for controlled experiments are typically induced in the laboratory or manipulated in the field. In contrast, subjects' beliefs tend to be constructed from the context provided in single-shot experiments or from prior outcomes in multiple-round experiments. Experimentalists have used various methods for measuring beliefs, especially in experiments that pertain to Bayes' rule or alternative heuristics for combining different sources of information (Grether 1980, 1992). Even when the primary focus is not on beliefs, it is often the case that direct elicitation procedures can provide useful insights for understanding the divergence between observed behavior and theoretical predictions, e.g., formation of beliefs about other players' decisions in games where equilibrium theories provide clear predictions about strategies and beliefs (Nyarko and Schotter 2002). Understanding the connection between beliefs and decisions can help discriminate between classes of theories, e.g., "rational" voting theories based on beliefs about the chances of affecting election outcomes versus noninstrumental theories based on expressive or other social motivations. In any situation involving risk, individual beliefs are a

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crucial aspect of the theory underlying decision-making, and accurate elicitation of beliefs can be extremely useful.

Financial incentives can be used to elicit beliefs about the probability of an event by letting subjects choose between alternative random lotteries for which the payoffs depend on whether the event occurs or not. One approach is to adjust the lottery payoffs to penalize larger differences between the elicited probability and observed outcome, so that the number of possible payoff amounts equals the number of feasible probability reports. A second approach restricts payoffs to one of two possible levels, say \$0 and \$2, and lets subjects choose between event-based and random number-based lotteries. The most common procedure in this second class is the Becker-Degoot-Marschak (BDM) procedure, in which the random number lottery can be thought of as an offer price at which the event lottery would be sold if the offer exceeds the reported probability. This latter approach, with only two payoffs, does not require any assumptions about risk preferences since the utilities for the high and low payoffs can be normalized to 0 or 1 without loss of generality. The popular quadratic scoring rule uses the former, penalty-based approach, which requires an assumption of risk neutrality or secondary measures of risk preferences to extract a subjective probability from the subject's choice among alternative lotteries.

This paper proposes a binary-payoff elicitation mechanism in which the choices are presented as a two-stage menu with only two possible payoffs for each decision. Each row of the choice menu contains an "event lottery" that pays if the event occurs, and a row-specific random lottery with a specified exogenous probability of the high payoff. The subject expresses a preference between these lotteries in each row, and the "crossover point" in the second-stage, fine-grid menu determines a subjective belief as defined by Savage (1971). The synchronized nature of the payoffs between the event and random lotteries is designed to induce truthful reporting of beliefs regardless of risk preferences or aversion. This theoretical invariance to risk attitudes distinguishes the proposed procedure from some other proper scoring rules such as the quadratic scoring rule. In addition, the row-by-row comparisons made by subjects in the choice menu help make this procedure more transparent than the price-based Becker-Degroot-Marschak procedure. The performance of this synchronized two-stage lottery choice method is experimentally tested against commonly used alternative methods.

I. Belief Elicitation Methods

A. *Quadratic Scoring Rule*

The standard approach to eliciting a probability with incentives is the quadratic scoring rule (QSR), originally proposed by Brier (1950) in the context of weather forecasts phrased as probabilities. The intuition is that the forecaster is paid a fixed amount of money, which we normalize to 1, minus a quadratic penalty for placing high probability on an event that does *not* occur. Let the reported event probability be denoted by R, so the penalty is $(1-R)^2$ if the event occurs and R^2 otherwise. If the person's subjective probability of the event is denoted by P, then a person's expected utility would be $PU\left(1-(1-R)^2\right)+(1-P)U\left(1-R^2\right)$. An extremely

risk-averse person would tend to choose reports close to 0.5, which would equalize the payoffs for each event.[1] In contrast, a risk neutral person would maximize expected payoff, which is a quadratic function of the reported probability:

$$(1)\quad \text{Expected Payoff for QSR} = 1 - P(1-R)^2 - (1-P)R^2$$
$$= 1 - P + 2RP - R^2.$$

This expected payoff is maximized by setting $R = P$, i.e., by reporting "truthfully." Similarly, a logarithmic scoring rule would deduct a penalty calculated from the log of the probability reported for the event that is observed. Under risk neutrality, the resulting expected payoff function is: $1 + (P)\log(R) + (1-P)\log(1-R)$, which is also maximized by reporting truthfully.[2] Rules that provide an incentive to reveal one's subjective probability are called "proper." This analysis, however, is based on an assumption of risk neutrality, which could necessitate a simultaneous elicitation of risk attitudes (Holt 1986).

Although the QSR only theoretically results in truthful elicitation under conditions of risk neutrality, it has been widely used and has provided valuable insights in the laboratory and the field. For example, Duffy and Tavits (2008) used a quadratic scoring rule to elicit beliefs about the probability that one's vote would be pivotal in the sense of making or breaking a tie. Subjects who placed a higher probability that their vote would be pivotal were more likely to vote. Elicited probabilities, however, were much closer to one-half than observed proportions of pivotal votes, which could be due to risk aversion bias, overconfidence or other factors.

Comparisons of alternative elicitation methods are relatively recent, with the exception of Huck and Weizsäcker (2002), who found that the quadratic scoring rule yielded more accurate predictions than a "bidding" mechanism that provided incentives for a risk-neutral individual to reveal a subjective probability.[3] The beliefs elicited with both procedures, however, tended to be biased toward the uniform prior (0.5).

Palfrey and Wang (2009) compare three different elicitation methods by asking subjects in an observer role to forecast behavior of other subjects who played a series of 2×2 matching pennies games, using a data series taken from Nyarko and

[1] Kadane and Winkler (1988) prove that the theoretical effect of risk aversion is to bias reported probabilities for the QSR toward one-half.

[2] Intuition suggests that the logarithmic rule would also exhibit a pull-to-center effect with risk aversion, since equal reported probabilities would tend to equalize payoffs whether or not the event occurs. A proof of this result is provided by Armantier and Treich (2013).

[3] The bidding mechanism was essentially a Becker-DeGroot-Marschak procedure for eliciting the *monetary value* of a lottery and using the resulting value to infer a probability. It involved asking subjects to bid for an event lottery that paid an amount, say 5, if the event of interest occurs and another amount, say 2, if the event does not occur. This procedure has the attractive property that the event lottery is a clear payoff-relevant manifestation of the underlying belief. The maximum willingness to pay for this event lottery is elicited with a second-price auction procedure in which another bid is randomly drawn from the relevant interval. If the randomly generated bid turns out to be less than the subject's own bid, then the subject pays the randomly generated bid and receives the event lottery, which is played out. Otherwise the subject earns nothing. This second-price auction would, in theory, reveal a maximum willingness to pay for the event lottery, which *assuming risk neutrality*, could be used to recover the subject's probability that the event would occur. This risk neutrality assumption was used by Huck and Weizsäcker (2002) in interpreting the results of the bidding mechanism.

Schotter (2002). Two of the rules were proper (assuming risk neutrality), and the third rule was a linear rule that paid the reported probability of the observed event, which in theory would induce subjects to report a boundary probability of one for the more likely event. This was a difficult task since the actual choice proportions for the game used in the Nyarko and Schotter experiment tended to hover in the 40–60 percent range. The calibration regression coefficients of the elicited probability on the actual choices are significantly positive but somewhat flat at about 0.3 for two proper scoring rules, QSR and logarithmic. However, they are significantly *negative* for the linear rule, which tended to provide extreme and implausible probability assessments, as theory would predict for this improper rule. The assessments obtained with the quadratic rule were somewhat *more extreme* than those provided with the logarithmic rule, but the overall reliability of the two proper rules was comparable. The logarithmic rule, however, requires that probability assessments be constrained away from 0 to avoid infinite negative payoffs, which may reduce its attractiveness in some contexts.[4,5]

One way in which scoring rules can be derailed is when the subject has a "stake" in the outcome of the event being predicted, or when the subject can take an action to hedge against uncertainty associated with the outcome. The intuition here is that event-contingent payoffs due to forecaster stakes or hedging behavior (endogenous stakes) can interact with risk aversion (Kadane and Winkler 1988). As a result, experimentalists have tended to use scaled-down payoffs in elicitation procedures, which could tend to decrease the effects of non-linear utility and forecaster stakes.[6] Armantier and Treich (2013) conducted an elicitation experiment with payoffs scaled from hypothetical (0x) to low (1x) to high (10x) to assess payoff scale effects, and with other treatments designed to assess forecaster stakes and hedging effects. They used a quadratic scoring rule to predict events determined by throws of a pair of dice. In a graph with the objective probability on the horizontal axis, a line showing forecast averages is clearly too flat *even with low* (1x) *payoffs*, and the pull-to-center tendency is much stronger with high (10x) payoffs.[7] This "regression bias" is eliminated by using hypothetical payoffs for the QSR, where the forecast averages track the 45-degree line in a graph with objective probabilities on the horizontal axis. Nevertheless, the forecasts in the hypothetical payoff treatment exhibit

[4] To ensure comparability across scoring rules, Palfrey and Wang (2009) constrained all probability assessments to be at least 0.1 with all three rules used.

[5] Interestingly, the Nyarko and Schotter (2002) study had players forecast other players' behavior using a QSR. The forecasts made by these players (who had a "stake" in the outcome) were *worse* than those of the observers in the Palfrey and Wang (2009) study. In fact, there was essentially no correlation (calibration regression coefficient of about 0.03) between players' forecasts and actual decisions made by their partners in the game. The stated beliefs provided by the Nyarko and Schotter (2002) subjects did provide a better guide to the subject's *own decisions* than could be obtained from inferred beliefs that were estimated with a structural learning model.

[6] Scaled-down payoffs will tend to reduce risk aversion effects under conditions of increasing relative risk aversion (more risk aversion at scaled-up payoffs), but not otherwise, as Armantier and Treich (2013) point out. There is considerable evidence from laboratory experiments for the presence of increasing relative risk aversion (Binswanger 1980 and Holt and Laury 2002).

[7] Armantier and Treich (2013) report similar increases in flatness of forecast response lines in other treatments that manipulate the forecaster's stake in the event outcome or the opportunity to make hedging decisions. Blanco et al. (2010) also provide evidence of confounds from hedging when incentives to hedge are relatively strong and transparent. Such incentives are removed by using a random device to determine whether a subject's payoff will be determined by stated beliefs or by the decision in the game (Blanco et al. 2010).

more noise, in the sense of systematically higher numbers of extreme responses (close to 0 or 1), responses on the wrong side of a half, and responses that reverse underlying objective probabilities. An interesting aspect of the Armantier and Treich incentivized procedures is that they selected only 1 of 30 elicitation tasks ex post to determine a subject's earnings. When a single decision is used, the payoff scale is set high enough to allow for the possibility of a reasonable earnings level, which tends to enhance risk, and hence, could strengthen any pull-to-center bias associated with a combination of risk aversion and high stakes.

One key procedural choice for the QSR is how to label the decisions. Some studies include probability labels and specifically encourage reporting of "true beliefs" (e.g., Nyarko and Schotter 2002) with hints about expected payoff maximization. For example, Offerman, Sonnemans, and Schram (2001) instruct subjects that "your expected PROBABILITY-PAYOFF is maximized if you report probabilities truthfully." More recent studies tend to employ neutral decision terminology with guidance that does not suggest risk neutrality (e.g., Holland, Massoni, and Vergnaud 2010 and Armantier and Treich 2013). It would be more problematic to encourage revelation of true beliefs in risky high stakes settings with one randomly selected decision used for actual payoffs. Another procedural choice to be made is how to present the QSR incentives. Some researchers provide subjects with the mathematical formula, as done by Nyarko and Schotter and others, which could be difficult for subjects unless they are encouraged to report true beliefs, as is typically done when the incentives are explained in terms of the penalty formula. An alternative is to provide a table with one row for each possible report, as done by McKelvey and Page (1990) and many others since, e.g., Armantier and Treich (2013). Rows of the payoff table could be either labeled with "chances in 100" probability labels or neutral numbered labels. Trautmann and van de Kuilen (2014) conduct one treatment with probability labels and explicit mention of the optimality of truthful reporting and one without labels or mention of payoff maximization and find no significant difference between the two treatments in terms of the likelihood whereby elicited probabilities for complementary events add to 1.

Any theoretical risk aversion or risk preference bias with QSR could potentially be corrected. Offerman et al. (2009) present a test of such a debiasing procedure that involves obtaining risk preference measures that are targeted to be relevant to the QSR decision problem. This correction is somewhat complicated, however, since it involves ten additional choice menus that are used to estimate probability weighting and utility curvature parameters. In any situation where risk preferences are separately measured and then used to correct QSR reports, several additional assumptions are required and measurement error can be compounded. Experimenters often assume a particular functional form for the utility function (and sometimes even a probability weighting function). For example, Fiore et al. (2009) used a Holt/Laury (2002) choice menu to estimate coefficients of constant relative risk aversion. When these estimated parameters are then used to correct QSR belief reports, a combination of specification error, measurement error, and instability of risk preferences can potentially complicate the situation. Finally, the measurement of risk preference is sometimes done in conjunction with the ex post selection of a single task, out of many, to use for actual payment (as a control for wealth effects), which would tend

to enhance the payoff risk and the effects of risk aversion or preference that need to be debiased. Offerman et al. (2009) find a greater need for risk aversion corrections to QSR reports in their treatment with payment for a single, randomly selected decision, as compared with their "pay all" treatment.[8]

B. *Binary Payoffs and the Becker-DeGroot-Marschak Mechanism*

An alternative to wrestling with risk preference issues is to use binary payoff procedures, e.g., with payoffs of $0 or $1, which would not be affected by risk attitudes or other curvature properties of a utility function, an approach taken to induce risk neutrality in other contexts (Roth and Malouf 1979). In fact, Grether (1978) pursued this avenue, by asking subjects which event they considered to be more likely, with a fixed money payment to be made if the response matched the actual event. This procedure provides a relative ranking of probabilities instead of a precise probability. In theory, it is possible to elicit an event probability by using any proper scoring rule with payoffs that are essentially tickets for a lottery involving a fixed prize amount.[9] McKelvey and Page (1990) transformed QSR payoffs into lottery tickets in this manner, but they still observed that subjects tended to report a "convex combination of the prior and the correct posterior," which shows up as a line that is too flat relative to the 45 degree prediction line in a graph with the correct Bayes' prediction on the horizontal axis (their Figures 2a and 2b). This binary lottery method of inducing risk neutrality is also taken in a couple of recent papers. Hossain and Okui (2013) use a Bayesian processing task to evaluate the QSR with and without the binary lottery. Prediction errors are lower with the binary lottery procedure, with most of the difference being observed for subjects who exhibited risk aversion in a separate task. Harrison, Martínez-Correa, and Swarthout (2015) also implemented a Bayesian information processing task. The proportion of red balls in a bingo cage was randomly determined behind a blinder, with all proportions being equally likely, before subjects were allowed to view the cage as it was spun ten times. Here the sample information (the subject's visual impression of the proportion of red balls) is nonnumeric, so is not possible to derive a posterior probability without additional modeling assumptions. The authors used the actual proportion of red balls in the cage as a proxy for the underlying subjective probability. The average distance between the reported number of red balls and the actual number was about 15 in the money payoff treatment and about 13 in the binary lottery payoff treatment, which indicates (along with some other econometric analysis) that the binary lottery procedure improved the accuracy of the elicitation. Even with the binary lottery procedure, however, there is some indirect evidence of a pull-to-center bias. The actual numbers of red balls for the four treatments were 17, 51, 57, and 87, and the average QSR reports with binary lottery payments were 31, 54, 60, and

[8] This observation is consistent with the results reported in Laury (2012), who compared the numbers of "safe" choices made in a menu for measuring risk aversion. With low stakes, there was no difference in results for the "pay all" and "pay one" (randomly selected decision) treatments. But scaling payoffs up by a factor of ten and only paying one randomly selected decision out of ten resulted in a significant increase in the number of safe choices.

[9] It is also possible to use lottery ticket payoffs to elicit other features of subjective distributions, e.g., means, quantiles, and variances (Schlag and van der Weele 2013).

68, which suggests a pull-to-center for the two extreme treatments. Of course, the "flatness" observed by McKelvey and Page (1990) and Harrison, Martínez-Correa, and Swarthout (2015) could be due to a bias in Bayesian information processing or to a failure to induce risk neutrality, or to a mix of both. To date, experimental economists have been hesitant to use binary lottery procedures, despite their theoretical appeal, because of insufficient evidence that the extra lottery layer does what it is supposed to do. See, e.g., the Selten, Sadrieh, and Abbink (1999) paper with the provocative title: "Money Does Not Induce Risk Neutral Behavior, But Binary Lotteries Do Even Worse."[10]

A different approach based on binary payoffs was taken by Grether (1992), who adapted the value elicitation procedures used by Becker, DeGroot, and Marschak (1964) in order to elicit probabilities. The resulting "BDM probability pricing task" involves asking the subject to report a number, e.g., the chances in 100 of a particular event. The experimenter then generates a random number N between 0 and 100 to determine a "Number Bet" with N chances in 100 of paying a fixed amount, e.g., \$1. If N is greater than the subject's reported chances in 100, then the subject's payoff is determined by playing out the Number Bet (using a second random draw). But if N turns out to be lower than the assessed chances in 100 for the event, then the subject's payoff is \$1 if the event occurs, \$0 otherwise (think of this as the "Event Bet"). So the Event Bet is selected by the experimenter to be used for the subject if it provides a higher chance of receiving the \$1 payoff, and the Number Bet is selected otherwise. It is obviously in the subject's best interest to report truthfully, and the binary nature of the payoffs ensures that this incentive is unaffected by risk attitudes. In a nutshell, Grether's approach was to use the BDM insight to elicit a selling price *in probability terms*, which yields the lowest probability associated with the number bet that the subject is willing to accept in exchange for the event bet.

Grether's (1992) experiments were run with and without financial incentives, using bingo cages and transparent randomization procedures for choosing whether to make draws from one cage or another. In some of the sessions, Grether would ask subjects which cage was more likely to be the one being used, with a payment that depended on whether the subject's guess was correct. The same subjects were also asked to provide a probability assessment of the chances that one cage is being used, with the BDM elicitation procedure designed to induce truthful revelation. Behavior did respond to relevant information in a reasonable manner, but there was a strong "representativeness bias" when the sample of draws matched the proportions of one of the cages (which would often cause sample information to be weighted too heavily relative to prior information). A striking result of Grether's study is that there were three times as many "nonsense or incoherent responses" without monetary incentives, e.g., putting a higher probability on the cage previously designated as being *less* likely.

Other types of bias have also been observed with BDM. One possibility is that responses could be biased upward if the process is perceived to be analogous to the

[10] Prasnikar's (1993) assessment of the binary lottery procedure was more positive, but she reported performance problems with the procedure that were attributed difficulties that some subjects had in reducing compound lotteries.

selection of a selling price, i.e., a "willingness to accept" bias, but this is hopefully avoided by using neutral terminology that abstracts away from buying or selling. Grether's (1992) regressions did yield somewhat low correlations between actual probabilities and those elicited with the BDM procedure. Möbius et al. (2007) also used a BDM probability pricing procedure to study belief formation in a setting with binary priors and signals. Their procedure is cleverly explained to subjects in terms of whether to substitute a payoff based on the subject's own performance for a payoff based on the performance of a robot named Bob, who essentially embodies the number bet. They find considerable evidence of conservatism (inadequate responses to sample information relative to a Bayesian benchmark). Holt (2006) as well as Holt and Smith (2009) report some success with a BDM probability pricing procedure, although there was evidence of over-weighting of low prior probabilities.[11] Note that these biases may be caused by difficulties associated with probability perception or Bayesian information processing rather than by difficulties with the elicitation technique used. None of these BDM-based studies provided a comparison of alternative elicitation methods.

C. *Comparisons of QSR and BDM*

Hollard, Massoni, and Vergnaud (2010) were the first to provide a comparison of the QSR and the BDM probability pricing rule. Subjects in their experiment were asked to indicate probabilities about whether their own decision on a quiz or perceptual task is correct. The results are compared with probabilities obtained from an unincentivized "Free Rule" method of just asking for a stated confidence level. Unlike most prior studies, probability numbers or "chances in 100" terminology was not used for the QSR, which simply involved choosing among ordered but unlabeled rows in a table of paired payoffs for each event. All three approaches are compared in terms of the accuracy of stated or implied beliefs in predicting actual success on the cognitive task. The BDM method outperforms the other two methods, despite the complexity involved with reporting beliefs and substituting another lottery derived from a randomly generated success probability if it yields a higher probability than the reported beliefs. Even the simple unincentivized Free Rule outperforms the Quadratic Scoring Rule with unlabeled decision rows, which yields extreme (0/1) or uninformed (50/50) responses about two-thirds of the time (Hollard, Massoni, and Vergnaud 2010). So unlabeled QSR incentives alone are not as effective as unincentivized elicitation in this context, but this does not mean that the QSR with probability "chances in 100" labels would not exhibit improved performance.

Trautmann and van de Kuilen (2014) provide a comparison of a BDM value elicitation procedure (with and without corrections for risk aversion), a menu-based

[11] Hurley and Shogren (2005) find a similar data pattern of overestimating low probabilities and underestimating high probabilities in the context of a Bayes' rule experiment. However, instead of eliciting subjective probabilities with BDM, Hurley and Shogren (2005) elicit predictions about the number of red chips that will be drawn (with replacement) from the can with known numbers of red and white chips. They note that the biased results indicate that there may be an error in the induction of beliefs or in the elicitation of beliefs, without being able to distinguish between those errors.

BDM probability pricing procedure, and QSR (with and without corrections to eliminate risk aversion and probability weighting bias). They find that the corrections did not increase the accuracy of responses (elicited probabilities of other players' decisions in an ultimatum game). In this context, however, there was not much difference between the performance properties of various incentive-based elicitation procedures that were used. In addition, these incentivized procedures did not perform any better than unincentivized introspection of other players' decisions in an ultimatum game in terms of accuracy and additivity (i.e., the likelihood with which elicited probabilities of complementary events sum to 1).[12] This is a striking result: just asking for a belief probability with no incentives does as well as complicated rules with nice theoretical properties or clever corrections for risk attitudes. Although it appears that unincentivized elicitation can yield adequate results in some contexts, there are many reports of diminished accuracy and a higher incidence of errors and nonsensical results when incentives are not used, e.g., Grether (1992); Armantier and Treich (2013); Hollard, Massoni, and Vergnaud (2010); and Gächter and Renner (2010). In the absence of a reliable set of criteria to determine when incentives matter and when they do not, it seems prudent to use incentives, together with an elicitation procedure with binary payoffs that is not distorted by such incentives.

D. *Efforts to Improve the BDM Procedure*

A major problem with the standard BDM probability pricing task is the abstraction associated with thinking about reporting a chances-in-100 number that, together with a randomly determined number, N, determines whether the subject's payoff is based on an "event lottery" (which provides a high payoff if the event is observed) or an "N lottery" (a subsequent random process that provides the high payoff with probability $N/100$). Thus, it can be beneficial to present the BDM mechanism in a manner that is easier for subjects to understand. Recently, Hao and Houser (2012) tested a procedure based on an English clock auction (proposed by Karni 2009). Subjects with endowed beliefs were asked to choose when to exit an auction for an event lottery when competing against a dummy bidder that exits according to the likelihood of success in a random lottery. A subject who believes that the event will occur with low probability will exit the auction early and receive the random lottery. Conversely, a subject who believes the event will occur with high probability will stay in the auction past the exit of the dummy bidder and receive the event lottery. The reported beliefs implied by the subjects' exit points in the clock auction tend to match endowed beliefs more closely than beliefs obtained with a "declarative auction," which is essentially a direct implementation of the BDM method as a second-price sealed auction. Due to the nature of a clock auction, however, the subject's beliefs were not precisely elicited in a significant proportion of cases, i.e., those in which the dummy bidder exited the auction first.

[12] Trautmann and van de Kuilen (2014) did find that incentivized procedures were better at predicting an individual's own behavior than introspection.

In this paper, we propose and test an approach to eliciting beliefs that takes advantage of the theoretical properties of BDM probability pricing (e.g., independence of risk attitudes), but attempts to make the "pricing" task of stating a probability cutoff more concrete by presenting specific comparisons in an ordered manner. The idea is to use a discrete choice menu to implement the intuition from Savage (1971) and Kadene and Winkler (1988) that the subjective probability of an event is defined to be the number p for which a person is indifferent between getting paid \$1 if the event occurs and getting \$1 with probability p. The two-stage synchronized choice menu is intended to clarify incentives to subjects and make the elicitation procedure more transparent and understandable. Choice menus are commonly used for the elicitation of risk preferences, e.g., Binswanger (1980), Holt and Laury (2002), and are also often used for the quadratic scoring rules and for finding payoff or probability indifference points in other elicitation procedures (Trautmann and van de Kuilen 2014). A complicated menu with lots of rows may tend to yield extreme or focal (50/50) responses, and therefore, we adopt a two-stage approach in which the subject first makes choices between an Event Lottery and a coarse grid of Random Lottery options, and then a more refined set of choices is presented at the crossover point for the coarse grid.[13]

We are not the first to use a structured set of lottery choices to infer subjective beliefs from binary choices. For example, Fiore et al. (2009) present subjects with a menu of choices between a risky prospect (framed as damage from a fire) and an abatement option that reduces the probability of damage at a cost that changes in successive rows. The objective is to elicit the subjective probability of damage, both with and without the abatement option. There are four different money payoffs in each row, two for each option, which depend on whether damage occurs for the option (abate or not) that is selected. With four different payoffs in each row, the choices for any given subjective probabilities will be sensitive to unobserved risk preferences.[14] Therefore, the authors present subjects with a separate Holt and Laury (2002) choice menu (with exogenous probabilities) that is used to elicit risk preferences, which are then used to "back out" the subjective probabilities from decisions made in the various rows of the abatement purchase menu. In contrast, the choice menus used in our paper only have two money payoffs in each row. As a consequence, the elicitation of a "probability equivalent" crossover point can be used to estimate a person's subjective belief in a manner that is understandable and yet does not require a separate procedure for estimating and removing effects of nonlinear risk preference.[15]

[13] See Tversky and Kahneman (1992) for an early example of the use of a second-stage menu with refined choices, used in a procedure to elicit certainty equivalents. Anderson et al. (2006) used a two-stage version of the Holt and Laury (2002) procedure to measure risk aversion. There is another strand of the literature that dynamically adjusts the choices seen in later stages in response to earlier choices, e.g., the Becker, DeGroot, and Marschak (1964) estimation of utility. See Wang, Filiba, and Camerer (2010) for an optimized dynamic procedure that is used to estimate risk and loss aversion.

[14] Antoniou et al. (2015) employ a similar choice menu framed instead in the context of a betting task to elicit beliefs. However, as in Fiore et al. (2009), there are more than two potential payoffs in each row, requiring additional measurement of and adjustment for risk aversion.

[15] An alternative approach is to use a series of one-on-one subject-experimenter interactions to find a point of probability indifference, as was done in Abdellaoui, Vossmann, and Weber (2005).

In addition, some tests for ambiguity aversion have been based on probability indifference questions with two possible payoffs (which precludes the need for risk aversion corrections).[16] In a similar fashion, mechanisms to elicit "uncertainty equivalents" such as those in Andreoni and Sprenger (2012) also employ a menu-based procedure to recover the probability of a defined high payoff in a second lottery that induces indifference with an initial known lottery.[17]

The proposed two-stage lottery choice menu is designed to minimize some of the concerns with existing methods. It is intended to be more understandable and less frustrating to subjects and experimenters than the BDM pricing procedure. The two-stage procedure allows the researcher to begin with a coarse grid that lets subjects see the big picture, with more precise probability estimates provided in the second stage. Such precision is especially important when dealing with probabilities, which may not be well approximated by midpoints of coarse intervals. In addition, because the two payoffs are synchronized throughout the rows of the choice menu, individuals have an incentive to report subjective beliefs truthfully regardless of their risk attitudes. The fact that there is no bias from unobserved risk preferences eliminates the need to account and adjust for utility curvature as is often done with QSR either by using a separate mechanism to measure risk preference or by making payments in lottery tickets. This is a significant advantage, as measurement of risk attitudes can require additional assumptions and create a second level of potential noise and measurement error.

In concurrent independent research, two other papers also employ a two-payoff approach to elicit subjective probabilities without the need to correct for risk aversion. Andreoni and Sanchez (2013) use a menu with 21 rows to elicit a probability equivalent for the outcome of a discrete trust game in which a player receives 1 of 2 possible money amounts ($5 or $15). The crossover row in the choice menu (approximately) determines the subject's belief that they will receive the higher payoff amount. Since only two payoff amounts were used, the inferred beliefs do not depend on assumptions about risk aversion. The elicitation worked well in terms of returning reasonable beliefs in settings for which the experimenter could gauge the

[16] An early example is Kahn and Sarin (1988), who describe one scenario to subjects: "There are two urns containing red and black balls. A ball will be chosen randomly from one of the urns; if the ball is red, you will win $100. Urn 1 has an unknown proportion of red and black balls totaling 200. How many red balls (total of red and black balls equals 200) would you put in Urn 2 to make you indifferent between the two urns." Here the range is from 0 to 200, so an answer below 100 for Urn 2 (with a known proportion) would indicate ambiguity aversion. In their experiment, this range of possible proportions of red balls was varied, and the degree of ambiguity aversion was significantly related to the range. More recently, Baillon, Cabantous, and Wakker (2012) and Dimmock, Kouwenberg, and Wakker (2013) also use probability indifference to measure ambiguity aversion. In this literature, the probability indifference point is typically not thought of as a *belief* about the proportion of red balls in the urn with unknown proportions, but rather, it is thought of as a way to measure ambiguity aversion by comparing the indifference point with the midpoint of the range of possible proportions.

[17] Andreoni and Sprenger (2012) extract both certainty and uncertainty equivalents in an experiment and find that certainty equivalents reproduce the S-shaped probability weighting predicted by cumulative prospect theory (Tversky and Kahneman 1992) while uncertainty equivalents do not. Futhermore, on an individual level, they find that subjects who demonstrate a preference for certainty and a violation of first-order stochastic dominance through uncertainty equivalents are also more likely to demonstrate an S-shaped probability weighting based on certainty equivalent reports. In an earlier paper, McCord and de Neufville (1986) investigate "lottery equivalents," which are similar to uncertainty equivalents but do not require that the high payoff in the second lottery be equal to the high payoff in the first. They discuss the practical use of lottery equivalents in identifying individual utility functions and they also test the procedure using a "structured, written questionnaire" (McCord and de Neufville 1986, 60).

accuracy of the beliefs. Similarly, Trautmann and van de Kuilen (2014) elicit beliefs about other players' decisions in an ultimatum bargaining game, using a single-stage menu with 21 choices to elicit a probability equivalent. As mentioned previously, they found little difference between probability matching elicitation and QSR-based approaches, and they conclude that un-incentivized reporting of beliefs performs as well as incentivized methods in all respects except for predictions about an individual's own behavior. Here, beliefs being elicited are not directly induced by the experiment procedures, but rather, are evaluated on the basis of whether probabilities sum to 1 and whether the beliefs are accurately related to observed choice proportions of other players. In addition, these papers are only able to recover beliefs in intervals of 0.05 due to the number of choices in the menus. Our two-stage procedure allows for more refined probability elicitation.

We find that the synchronized two-stage choice menu used in this paper performs well compared to both BDM and QSR (both with and without probability labels). The choice menu has the lowest average absolute deviation from Bayes' predictions across elicitation methods studied although this difference is only significant in comparison with QSR (without probability labels). Similar results are obtained when the analysis is restricted to rounds in which subjects were not required to make any Bayes' rule calculations and could rely solely on "endowed" beliefs. When the posterior is not equal to one-half, the lottery choice menu generates a significantly larger number of reported beliefs within $+$ or -1 chance in 100 of the Bayes' prediction than any of the alternative methods explored. In addition, the reported beliefs obtained in both of our quadratic scoring rule treatments are pulled toward the extremes in a noticeable way, contrary to some other experimental results supporting the pull-to-center bias that could be caused by risk aversion. This extremeness tendency for the QSR has been reported in some prior studies (Dominitz and Hung 2004; Nyarko and Schotter 2002; Palfrey and Wang 2009; Hollard, Massoni, and Vergnaud 2010), and the implication cautions against "correcting" QSR reported beliefs for risk aversion as, in our case, this correction would result in even more extremeness for the majority of risk-averse subjects.[18]

II. Procedures

A total of 8 sessions with 18 subjects each were run with undergraduate students at the University of Virginia. All sessions followed the same basic procedures explained below but differed in terms of the belief elicitation mechanism that was utilized. There were 2 sessions (36 subjects total) in each of 4 treatments that varied the elicitation mechanism. The three elicitation mechanisms investigated were the Becker-DeGroot-Marschak "*N*-Lottery" method (BDM), the quadratic

[18] In general, the pull to center effect observed in many previous papers is strong in cases where payoffs are scaled up, e.g., the 10x Armantier and Treich (2013) treatment that yielded a strong pull to center, as compared with their hypothetical QSR treatment that produced no systematic deviations from the 45 degree prediction line. This pull to center is also clear in studies like Harrison, Martínez-Correa, and Swarthout (2015), which, in an attempt to control for wealth effects, base payoffs on a single high-stakes ($0 or $50) choice, selected at random ex post (see discussion on page 7). Similarly, Offerman et al. (2009) observe a stronger pull to center (and hence more of a role for de-biasing) in their "pay one" treatment, as compared with their "pay all" treatment.

scoring rule (QSR with and without probability labels), and the lottery choice method (LC).[19]

In each session, subjects were informed about the computer-simulated draws from two possible virtual cups containing different numbers of red and blue marbles. At the beginning of each round, the computer would select one cup at random. The "Blue Cup" would be selected with a probability of one-half (i.e., if a computer-generated 6-sided die throw was a 1, 2, or 3) and the "Red Cup" would be selected with a probability of one-half (i.e., if a computer-generated 6-sided die throw was a 4, 5, or 6). The Red Cup contained 2 reds and 1 blue and the Blue Cup contained 2 blues and 1 red. After the cup was selected (but before subjects were informed of which cup was used), each participant was shown a series of draws with replacement from the selected cup. The number of draws ranged from 0 to 4 depending on the round with the sequence of draws in the 12 rounds being 0, 1, 2, 3, 4, 0, 0, 1, 2, 3, 4, 0. After any draws were shown to the subject, a mechanism was used to elicit each subject's beliefs about the likelihood that the Red Cup had been selected for that round. Then the cup used was revealed and payoffs were calculated for the round. Subjects earned amount equal to the sum of their payoffs from all 12 rounds. A common "seed" value was used to generate the same random series of cup selections and sample draws across sessions. We chose to use a common random seed to avoid the possibility that differences in realizations of cups or random draws may cause different behavior across treatments. When choosing a random seed, we strived to achieve an actual proportion of Red Cups selected close to the ex ante probability of Red Cup selection of 0.50 and conditional probabilities after one draw close to 0.67 as is predicted by Bayes' rule.[20] Note that, even with a single random seed used across sessions, each of the 18 individuals within a session saw different sequences of realizations. Thus, with 12 cup realizations and 20 draw realizations per person, each session of 18 subjects consisted of $18 \times 32 = 576$ random realizations, so there remains quite a lot of randomness within each session.

As mentioned previously, the BDM incentives can be difficult for subjects to understand, and indeed the standard way to use a BDM procedure is to just tell subjects that it is in their best interest to report their true belief. In the case of the BDM or LC treatments, there was no need to mention a risk neutrality qualification, and the instructions explained why it was in the subject's interest to make a decision that corresponds to their true beliefs.[21] For the case of the quadratic scoring rule, we included separate treatments that did or did not use "chances in 100" terminology. In the "QSR-chances" treatment, the rows in the choice table were labeled in terms of chances in 100 for the Red Cup, and the wording of the instructions contained

[19] Each participant was paid $6 for attending and any additional earnings from the experiment. Total earnings for the experiment, which lasted approximately one hour, averaged about $21 (including the $6 attendance fee) and were paid in cash to all participants at the conclusion of the experiment. Instructions were both shown on the computer monitor and read aloud, and subjects made decisions on isolated computer terminals. A printed version of the web-based instructions for all treatments can be found in the Appendix.

[20] Averaged across both treatments for all subjects in a session, the ex post unconditional probability of a Red Cup was 0.505, the ex post probability of a Red Cup given a red draw was 0.69, and the ex post probability of a Blue Cup given a blue draw was 0.645.

[21] For example, the final instructions page for the LC treatment reads: "Coming down the table, your choices should switch from the right to the left side at the point where the chances in 100 for the Random Lottery exceed the chances in 100 that you think the Red Cup is being used."

references to optimality in terms of expected payoffs.[22] This treatment was designed to give the quadratic scoring rule its best chance and to be consistent with optimality advice provided for the BDM and LC rules. In contrast, the QSR-numbers treatment consisted of a quadratic scoring rule with a more neutral frame, where the subject was asked to choose a numbered row in a table instead of reporting chances in 100. The instructions, patterned after Armantier and Treich (2013), stressed that there is no correct answer and that different individuals may choose different numbers. The instructions did not suggest that the row number should match the subject's beliefs about the chances that the event would occur.[23] In order to be parallel to the instructions for our other three treatments, we added a comment mentioning that it may not be profitable to choose a row number below 50 if the Red Cup was considered to be more likely, or to choose a number above 50 if the Blue Cup was considered to be more likely. This statement accurately describes incentives, regardless of risk attitudes.

In terms of implementation, the BDM treatment asked subjects to enter the "chances out of 100" that the Red Cup was selected in that round. Then, as was explained to subjects, a number between 0 and 100 would be randomly generated by the computer. If the random number was smaller than the reported beliefs, they would receive a payoff of \$2 if the Red Cup was actually used and \$0 otherwise. If the random number was larger than the reported beliefs, then a second random number would be generated and the subject would receive \$2 if the second random number was smaller than the first random number, and \$0 otherwise. As explained previously, this procedure encourages truthful revelation across all risk attitudes but can be more difficult to comprehend.

In both QSR treatments, subjects were shown a table with all possible choices and the payoffs associated with each choice depending on whether the Red Cup or Blue Cup was actually used. Those payoffs reflected the quadratic scoring rule where subjects were penalized in proportion to the square of the reported probability they assigned to the event that did *not* occur. For example, given a subject who reported X would receive $\$2 - \$2(X/100)^2$ if the Blue Cup was being used and $\$2 - \$2((100 - X)/100))^2$ if the Red Cup was being used. As noted previously, this procedure is widely used and appears to be relatively easy for subjects to understand.

The LC treatment is designed to test the two-stage lottery choice method of belief elicitation discussed in the previous section. It is essentially a structured choice menu that preserves the beneficial theoretical properties of the BDM method, but it is designed to be more intuitive and understandable to subjects by separating the decision into a structured series of binary comparisons in the rows of a table.[24]

[22] In this treatment, the instructions followed the common procedure (e.g., Nyarko and Schotter 2002; Offerman, Sonnemans, and Scrham 2001) of asserting that reporting one's true beliefs for the QSR will maximize expected payoffs, although this practice is not followed in some recent papers. Our QSR-chances instructions read: "If your decision represents your true beliefs about the chances that the red cup is being used, then your expected earnings (determined by multiplying probabilities times corresponding payoffs) will be maximized."

[23] The key paragraph of the Armantier and Treich (2013) instructions (translated from French) reads: "There is no correct or incorrect choice. In general, however, you may find it profitable to choose a higher choice number when you think the chances that the event occurs is higher."

[24] Such choice menus are commonly used to measure risk aversion, see the discussion of the literature in Holt and Laury (2014).

Table 1—Lottery Choice Menu: First Decision Table

Random Lottery			Red Cup Lottery
0 chances in 100 of $2.00	◘	◙	$2.00 if the Red Cup was used
10 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
20 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
30 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
40 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
50 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
60 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
70 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
80 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
90 chances in 100 of $2.00	◘	◘	$2.00 if the Red Cup was used
100 chances in 100 of $2.00	◙	◘	$2.00 if the Red Cup was used

Subjects were given a table, shown in Table 1, where each of the 10 rows represents a choice between a "Red Cup Lottery" that pays $2 if the Red Cup is selected and $0 otherwise, and a "Random Lottery" that pays $2 with varying chances in 100 (i.e., 0, 10, 20, 30, …, 100 down the rows of the table).

Subjects were asked to choose between the Red Cup Lottery and the Random Lottery in each of the ten rows. The decisions for this table essentially involve choosing which row to cross over from the Red Cup Lottery (which is the best choice when the Random Lottery promises a payoff with 0 chances in 100 at the top) to the Random Lottery (which is the best choice when the Random Lottery promises a payoff with 100 chances in 100 at the bottom). A person who crossed over more than once was forced to re-choose.[25] After a set of decisions with a single crossover was entered for this coarse grid table, the subject was asked to refine the decision on the crossover row using a drop-down select menu with rows for each increment in probability, e.g., 40, 41, … 50 chances in 100. Subjects were made aware that one of the rows of the first table (or potentially the refinement drop-down menu if the initial random number corresponded to the crossover row) would be selected at random to determine their earnings in accordance with the lottery chosen. If a subject had chosen the Red Cup Lottery on the randomly selected row, the payoff would be $2 if the Red Cup was selected and $0 otherwise. If, instead, a subject had chosen the Random Lottery for the selected row, a second random number would be generated to determine earnings. If the second random number was smaller than the "chances out of 100" for the selected Random Lottery, the subject would receive $2, and they would receive $0 otherwise.

[25] One consideration is whether the enforcement of a single crossover biases the data in favor of LC. There is nothing analogous to multiple crossovers in the QSR treatment (where subjects must choose a single row) or in the BDM treatment (where subjects must enter a single probability). Therefore, we do not believe that preventing multiple crossovers provides an advantage to the LC treatment. Our intention was to enhance an understanding of the incentives by forcing a single crossover and providing starting points that avoid a sure zero payoff in the top row and that ensure a safe $2 payoff in the bottom row. Since a single crossover was enforced, we do not have data on the number of initial crossovers made by an individual or the number of times they submitted a table with more than one crossover. An alternative would have been to allow multiple crossovers and discard the data, or use an averaging method as in Holt and Laury (2002). Discarding data for those who cross over multiple times would bias the results by excluding confused people for the LC treatment and not for the other treatments.

To better understand how the second stage works, suppose that the subject chose the Event Lottery in the top 6 rows and crossed where the Random Lottery provides 60 chances in 100 of the \$2 payoff. This means that the event lottery is thought to have fewer than 60 chances in 100 and more than 50 chances in 100 (based on the choice in the previous row). Thus, the second-stage drop-down menu would have choices of 50, 51, … 60. We were initially worried that a possible tendency to choose at the midpoint of the second stage would actually degrade the performance of the LC. For example, if the belief is at 50 (after no draws or balanced draws), then the midpoint choice would be at 55. This did not occur, since the percentage of correct choices is quite high for the LC treatment when the posterior is 0.5, as will be apparent from the discussion of results.[26]

It is interesting to compare the payoff incentives for the LC and BDM procedures with the QSR expected payoffs in equation (1). The LC menu with the second stage is essentially a list of 100 rows, indexed from $N = 1$ to 100, each with a "random lottery" on one side that provides N chances in 100 of a payoff, π, which was \$2 in the experiment. Since each row is equally likely to be used, think of N as being uniform between 0 and 100. Consider a continuous version of the LC with $U(0)$ normalized to 0, and let $n = N/100$ be uniform on [0, 1]. Then the expected utility for reporting a probability R can be expressed: $\int_0^R PU(\pi)\,dn + \int_R^1 nU(\pi)\,dn$, where the first term corresponds to the case where the random lottery is unattractive, $n < R$, and hence the subject prefers the event lottery that pays π with subjective probability P. The second integral corresponds to the case where n is high enough so that the person prefers the random lottery that pays π with probability n. By integrating and rearranging terms, this expected utility can be expressed as: $U(\pi)\left[1 + 2RP - R^2\right]/2$. Because payoffs are binary (either 0 or π), the utility term factors out of the expected utility (risk preferences do not matter), and the bracketed quadratic function in the expected utility expression only differs by an additive constant from the QSR payoff on the right side of equation (1) that requires an assumption of risk neutrality. This analysis also describes a BDM procedure where R is a number analogous to a selling price that the subject selects, and n is a randomly generated bid. If the bid is below R, the person "keeps" the event lottery that pays $\$\pi$ if the event occurs, and if the randomly generated bid is higher than the selling price, the person accepts the bid for a probability n of the π payoff.

In order to assess the incentive properties of different procedures, it is useful to consider expected payoffs when the subjective belief is different from 0.5 so that the QSR would bias the elicited probability toward 0.5 in the case of risk aversion and toward a more extreme value in the case of risk preference. Figure 1, for example, shows the expected utilities for various reported chances in 100 listed on the horizontal axis, when the person's subjective belief is 0.8 (80 chances in 100) that the event occurred. The gray line shows expected utility under QSR calculated from the payoffs in (1), scaled up by a factor of 2, and the dark line shows expected utilities

[26] In addition, the data graphs that follow do not reveal any systematic grouping of decisions in the middle of intervals for the LC procedure, as compared with the other procedures. In general, a tendency to select decisions in the middle of the ten-point intervals would not necessarily benefit the performance of LC since the majority of Bayes' predictions are close to the endpoints of those intervals (i.e., 11, 20, 50, 80, 89).

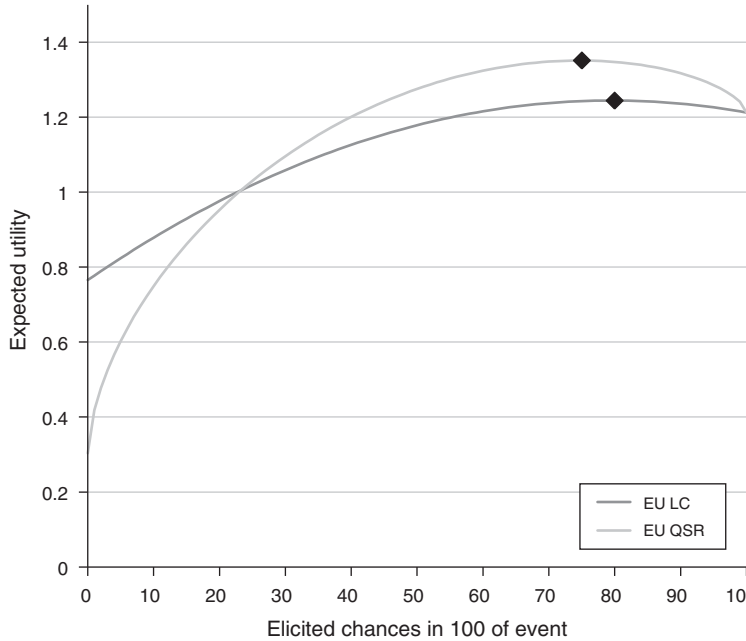

FIGURE 1. EXPECTED UTILITY PROFILES FOR A BELIEF OF 0.8 (80 *chances in* 100)

from the LC and BDM procedures with a payoff of \$2 as used in the experiment. The utility function used was $U(\pi) = \pi^{1-r}$, with a constant relative risk aversion parameter of $r = 0.4$ that is in a range commonly reported in many studies (Holt and Laury 2014). Note that expected utilities for all procedures are the same on the right side of the figure, when the reported chances are 100, since in each case reporting 100 provides the subject with the "event lottery" that provides $U(2)$ if the event occurs, i.e., with probability 0.8. With BDM, for example, a report of 100 results in a rejection of all "bids" and keeping the event lottery. For both the LC and BDM procedures, the expected utility is maximized at a report of 80, corresponding to the "true" probability. With the QSR and this moderate level of risk aversion, there is a downward bias in the sense that the expected utility is maximized at 75 (shown by the dark diamond mark on the gray line). In contrast, for a mild risk preference ($r = -0.3$) the expected utility under QSR would be maximized at 85, a bias toward the extreme. It is worth noting that neither of these biases are very large, and that both expected utility plots in the figure are fairly flat near the maximum, as would be expected from the quadratic nature of the incentives. These similarities suggest that differences between elicitation procedures may be largely due to transparency and changes in the thought process that the procedures induce. The lottery choice method was designed with these features in mind.

III. Results

Table 2 shows the median elicited belief across individuals for each treatment conditional on the Bayes' Rule prediction, which ranges from 6 to 94 depending on

Table 2—Median Reported Chances in 100 (*lower* 10 *percent*, *upper* 10 *percent*)

Posterior chances in 100	Lottery choice menu	BDM	QSR chances	QSR numbers
6	8 (4, 12)	**2** (0, 10)	**1** (0, 10)	**5** (1, 81)
11	23 (0, 50)	13 (1, 30)	**6** (0, 26)	**5** (0, 19)
20	20 (1, 50)	20 (0, 47)	**16** (0, 40)	**12** (0, 44)
33	33 (20, 50)	33 (0, 55)	33 (0, 50)	**27** (5, 50)
50	50 (41, 51)	50 (50, 70)	50 (50, 60)	50 (39, 73)
67	66 (50, 87)	66 (50, 83)	**70** (50, 93)	**70** (50, 95)
80	79 (50, 90)	79 (50, 100)	**90** (68, 100)	**81** (51, 100)
89	**90** (84, 100)	88 (60, 100)	**98** (85, 100)	**95** (56, 100)
94	77 (57, 99)	83 (60, 90)	**100** (68, 100)	**98** (92, 100)
Decisions within +/− 5 percent of Bayes' prediction not equal to 0.50	39%	25%	21%	13%

the observed draws. The numbers in parentheses at each Bayes' prediction denote the tenth and ninetieth percentiles of the distribution of observations. Notice that the median elicited beliefs are overall quite close to theoretical predictions, especially when the predictions are close to 50. There tends to be more variability at the extremes given the smaller numbers of observations with three or four matching sample draws. However, there does not appear to be any systematic overall bias across treatments in terms of median elicited beliefs. This is reassuring for the BDM treatment with neutral terminology, where the use of "selling price" terminology to explain the procedure might have produced an upward bias in the reported beliefs (selling prices).[27] The final row of Table 2 highlights the percentage of decisions within plus or minus 5 percent of the Bayes' rule prediction for all unbalanced draw rounds (i.e., where the Bayes' rule prediction is not equal to 0.50). This indicates that there is more concentration around Bayes' prediction with the lottery choice method when considering unbalanced draws.[28]

The closeness of the median elicited beliefs to the theoretical predictions for all treatments is not so surprising given (i) the prior probabilities were a half, which facilitates the information processing task, and (ii) prediction errors in opposite directions tend to cancel out. However, Table 2 does hint at a tendency for beliefs to be pulled toward the extremes and away from the center of 50 in the QSR treatments. The bold numbers in Table 2 indicate instances in which the median elicited belief is closer to the extreme (either 0 or 100) than predicted by theory. Notice that,

[27] The similarity of median predictions and Bayes' predictions in Table 2 is misleading, since there is a lot of dispersion around these medians, even though we used prior probabilities of a half. In order to assess whether the deviations for the experiment reported here are atypical, we identified two Bayes' rule experiments using the QSR with priors of 0.5 and probability labels on decisions: McKelvey and Page (1990) and Hossain and Okui (2013). The deviations from Bayes' predictions for our QSR-chances treatment are only slightly lower than the average for the McKelvey and Page data (an average deviation of 8.6 from our Figure 3 below, versus an average deviation of 9.7 that we estimated from their Figure 2a). Moreover, the average squared deviation from Bayes' predictions of 211.3 for our QSR-chances treatment is only slightly higher than the average squared difference of 170.1 reported by Hossain and Okui (2013). There were, of course, numerous procedural differences between these papers, but the overall similarity of deviation measures suggests that the results reported on this paper are not obviously out of line with some previous studies.

[28] If all draws are considered, the percentage of decisions within plus or minus 5 percent of the Bayes' rule prediction is 55 percent for LC, 47 percent for BDM, 46 percent for QSR-chances, and 38 percent for QSR-numbers.

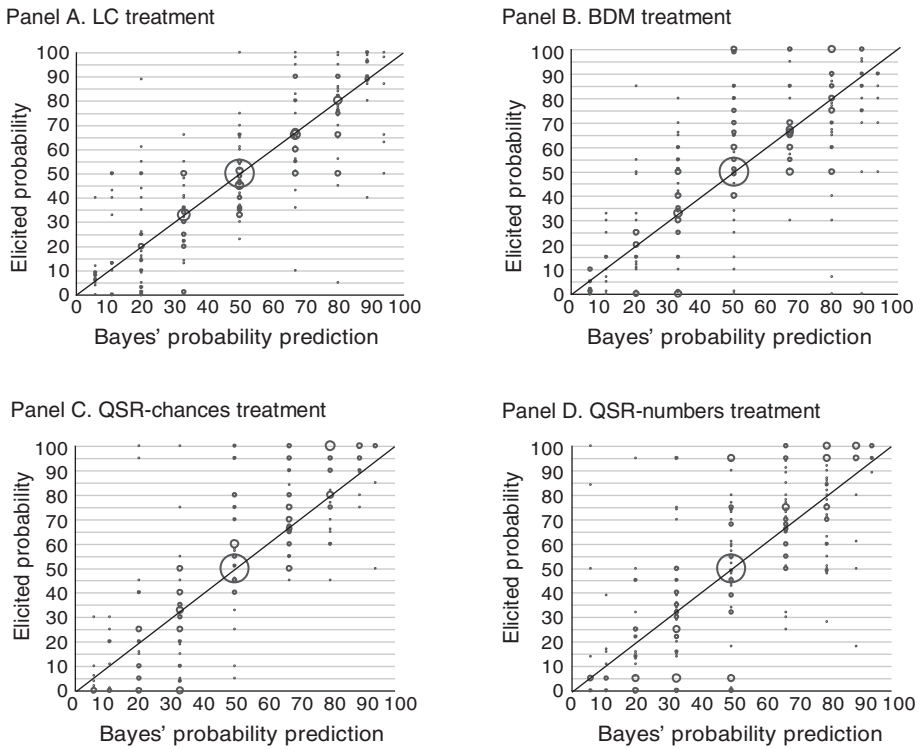

Figure 2. Bubble Graphs of Individual Decisions by Treatment (*all rounds*)

when the prediction differs from 50, all of the median beliefs with QSR-chances and QSR-numbers are highlighted with only 1 exception. Also, the tenth percentile is always 0 when the posterior is less than 50 and the ninetieth percentile is frequently 100 in the QSR-chances treatment. The observed "extremeness" with the QSR treatments is a feature of the data that will be discussed more fully later.

Still more information can be gleaned from examining not only the medians but the pattern of prediction errors for each treatment, which can be seen in Figure 2 panels A–D, where the bubbles are sized to indicate the frequency of decisions at that point (i.e., larger bubbles correspond to probabilities that were reported more frequently for a given Bayes' prediction). Looking at Figure 2, panel A, which depicts the LC treatment, note that reported probabilities with the largest frequencies tend to correspond to the Bayes' prediction (on the 45 degree line). Recall the percentage of decisions within plus or minus 5 percent of the Bayes' rule prediction is largest for LC, as shown previously in Table 2. This concentration on the prediction line is not observed for the other three treatments. Also, there appears to be noticeably noisier decision-making in the QSR-numbers treatment in Figure 2, panel D.

The data shown in Figure 2, panels A–D suggest that deviations will exhibit a clear pattern across treatments. Figure 3 shows the average absolute deviation by treatment for all rounds (light gray bars), rounds with either zero or balanced draws (dark gray bars), and rounds with unbalanced draws (black bars). Note that the three elicitation methods that are framed in probability terms generally perform

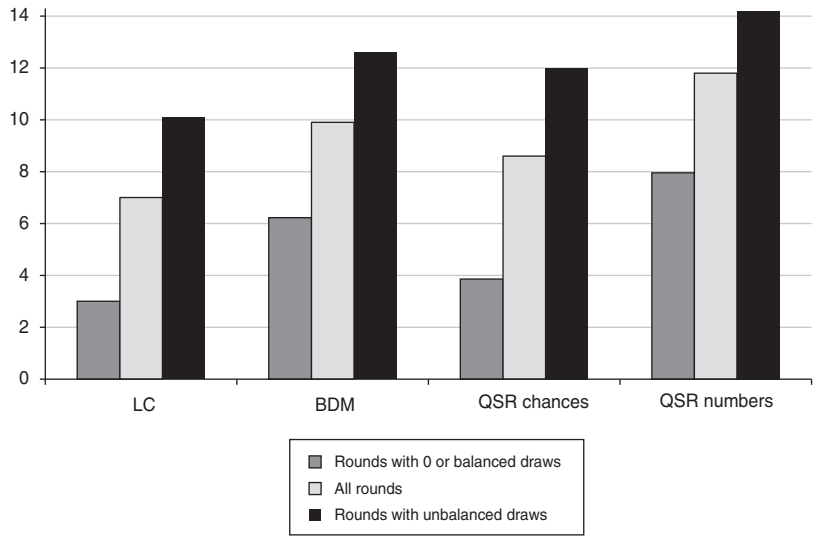

FIGURE 3. AVERAGE ABSOLUTE DEVIATION FROM BAYES' PREDICTIONS

better than the neutral QSR-numbers treatment. The "all rounds" light gray bars in Figure 3 illustrate our first result:

RESULT 1: *The Lottery Choice method yields lower average absolute deviations from Bayes' Rule predictions than all other mechanisms explored in this paper. However, this difference is only significant when Lottery Choice is compared to the Quadratic Scoring Rule method with neutral terminology where probability labels are not used* (*QSR-numbers*).

Support.—If we look at the overall treatment average across individuals for all rounds (the light gray bars), we find that LC has an average absolute deviation of 7.0 compared with 8.6 for QSR-chances, 9.9 for BDM, and 11.8 for QSR-numbers. Using a two-tailed randomization test across individual averages, the average absolute deviation with LC is almost significantly different from BDM (p-value $= 0.1004$) and significantly different from QSR-numbers at the 1 percent level (p-value $= 0.0081$).[29,30] The difference between QSR-chances and both LC and BDM is not significant but it is noteworthy that the two QSR treatments are significantly different from one another at the 10 percent level (p-value $= 0.0545$). Thus, providing the probability labels to the QSR choices significantly alters reported probabilities towards the Bayes' Rule prediction.

[29] We used 500,000 simulations for this randomization test and for all of the randomization tests reported throughout the paper.

[30] The nonparametric randomization tests reported here take all decisions by an individual and average them to get a single data point for each individual. An alternative approach would be to use all decisions in an OLS regression and generate robust standard errors through clustering by individual, a process similar to that used in Hossain and Okui (2013) to explain negative squared error. In this spirit, we ran two regressions (all rounds and

Considering all rounds of this experiment includes some rounds where there are unbalanced draws (i.e., one red and two blues, etc.), for which there are two separate aspects combining to make up an individual's reported probability. First, the individual must calculate a subjective probability, which may or may not be accurate according to Bayes' rule. Then the subject must translate that subjective probability into a reported probability given the elicitation mechanism. The desirable attribute for a belief elicitation mechanism is for the subject to truthfully report their subjective probability (regardless of the accuracy of that subjective belief in terms of its conformity with Bayes' rule).[31] When calculations are required to form subjective beliefs, it can be harder to assess whether an inaccurate (according to Bayes' rule) reported probability is due to a biased elicitation mechanism or simply a calculation error on the part of the subject. Thus, it can be informative to consider those rounds where no calculation was required by the subject to form beliefs. This situation occurs in rounds where there are 0 draws because all they have to rely on are the priors of one-half for each cup, which were explained in the instructions in the context of throwing a die and selecting the Blue Cup if the die throw is 1, 2, or 3 and selecting the Red Cup if the die throw is 4, 5, or 6. Similarly, little calculation is required when there are draws that balance each other out, where the numbers of red and blue marbles drawn is the same (i.e., one red and one blue or two reds and two blues). In these balanced situations, the draws essentially provide no additional insight into which cup was selected, so subjects are again forced to rely on the given priors of one-half. The dark gray bars in Figure 3, which show the average absolute deviation across all individuals in each treatment for rounds with zero or balanced draws, are lower than the all-rounds measures. This figure highlights our second result:

RESULT 2: *The differences in average absolute deviations detailed in Result* 1 *continue to be present in the absence of complicated Bayes' Rule processing.*

Support.—For rounds with no draws or balanced draws, which are shown by the dark gray bars in Figure 3, LC continues to have the lowest average absolute deviation. Consistent with the analysis on all rounds, two-tailed randomization tests

round with zero or balanced draws) of treatment dummy variables on the absolute deviation from Bayes' prediction. The coefficient estimates and p-values are similar and slightly lower in some cases as indicated in the table below:

	Constant	BDM	QSR-chances	QSR-numbers
Absolute deviation (all rounds)	7.01***	2.86*	1.56	4.78***
	($p = 0.000$)	($p = 0.094$)	($p = 0.229$)	($p = 0.007$)
Absolute deviation (0 or balanced rounds)	2.99***	3.26*	0.875	5.34**
	($p = 0.002$)	($p = 0.090$)	($p = 0.418$)	($p = 0.023$)

We get identical coefficients and significance levels (with slight differences in standard errors) if we run a treatment by treatment comparison between LC and each of these other mechanisms. The only other significant result in the treatment by treatment comparison is that the absolute deviation for QSR-numbers is significantly greater (with coefficients on the QSR-numbers dummy of 3.22 and 4.46) than QSR-chances at the 10 percent level.

[31] While it would be optimal to test if elicited beliefs matched "true" subjective beliefs, the latter is essentially unknown in most any context and so we view the Bayes' prediction as the most useful benchmark. However, if one could design a game where there was little possibility for other-regarding preferences or risk attitudes to play a role, a good elicitation mechanism should elicit beliefs that justify decisions. This is an interesting avenue for future research on belief elicitation.

on individual averages confirm that the difference between LC and BDM for rounds with zero or balanced draws is again almost significant (p-value $= 0.1064$) and the difference between LC and QSR-numbers is significant at the 5 percent level (p-value $= 0.0258$). In addition, the difference between the QSR treatments continues to hold as well, whereby the average absolute deviation for QSR-chances is significantly lower at the 10 percent level (p-value $= 0.0715$) than for QSR-numbers.[32] There is still no significant difference in average absolute deviation between QSR-chances and both LC and BDM. LC continues to have the lowest observed average absolute deviations if attention is restricted to only rounds with unbalanced draws, as illustrated by the black bars in Figure 3, but the only significant difference with unbalanced draws is between LC and QSR-numbers at the 5 percent level (p-value $= 0.0409$). This may be due to the heightened importance of more complex computations in rounds with unbalanced draws. While the Bayesian predicted probability is the best benchmark available to us, it is more likely in these cases that the subjective probability being elicited is not exactly equal to the Bayesian predicted probability (although any bias in subjective probabilities should be present across all treatments).

The next step in the analysis is to examine the nature of the prediction errors across treatments. As mentioned before, note that as Bayes' rule predictions diverge from one-half, the QSR treatment medians in Table 2 are actually pulled somewhat further away from the center towards 0 and 100. This can also be seen in Figure 2 panels C and D whereby larger bubbles, representing higher frequencies of decisions, are located above the 45 degree line for higher Bayes' predictions and below the 45 degree line for lower Bayes' predictions. Some of this pull is due to boundary decisions, which are never optimal. The proportion of boundary decisions (i.e., 0 or 100) was 10 percent in BDM, 15 percent in QSR-Chances, and 9 percent in QSR-Numbers, as compared to only 3 percent in LC. This represents our third result:

RESULT 3: *The lottery choice method has a lower incidence of suboptimal boundary decisions.*

Support.—The proportion of boundary decisions in the LC treatment is significantly different than QSR-chances at the 1 percent level (p-value < 0.001), BDM at the 5 percent level (p-value $= 0.0462$), and QSR-numbers at the 10 percent level (p-value $= 0.0530$) using a two-tailed randomization test on individual proportions of boundary decisions. Furthermore, this result is not driven solely by a small number of subjects repeatedly making extreme boundary decisions. Instead, the percentage of subjects who make boundary decisions at least once was significantly higher with QSR-chances (61 percent) than LC (14 percent) at the 1 percent level (p-value < 0.0001), QSR-numbers (33 percent), at the 5 percent level (p-value $= 0.0329$), and BDM (39 percent) at the 10 percent level

[32] Our conclusions are unchanged if we restrict attention to only those rounds with zero draws (which excludes the rounds with balanced draws). In fact, using only rounds with 0 draws, some results are more significant with the difference between LC and QSR-numbers now significant at the 2 percent level and the difference between QSR-chances and QSR-numbers now significant at the 5 percent level. The difference between LC and BDM continues to be almost significant with a p-value of 0.1060.

(p-value $= 0.0983$) using a Fisher exact test. In addition, the percentage of subjects making boundary decisions at least once was also significantly lower with LC than with both BDM (at the 5 percent level with a p-value $= 0.0309$) and QSR-numbers (at the 10 percent level with a p-value $= 0.0942$) again using the Fisher exact test. The somewhat surprising frequency of observed boundary decisions for the QSR-chances treatments occurred even though the payoffs are extremely flat toward the extremes and these boundary decisions were not observed nearly as often in the QSR-numbers treatment without probability labels.[33]

Another perspective of extremeness can be obtained by considering the frequency and directions of deviations from Bayes' predictions. Table 3 shows the average percentage (across individual subject percentages) of correct decisions,[34] deviations toward the center, deviations toward the boundary, and decisions on the wrong side of one-half. The percentages are for all rounds for which the Bayes' rule prediction does not equal one-half. The average across individual percentages for each category was used in order to maintain independence of observations used for statistical testing. Table 3 illustrates our fourth and fifth results:

RESULT 4: *The Lottery Choice Method yields significantly more reports consistent with Bayes' Rule predictions than all other mechanism explored in this paper.*

RESULT 5: *Both QSR treatments exhibit a strong pull toward the extremes not present in other mechanisms.*

Support.—First, note that the average individual proportion of correct decisions in the left column is much higher in the LC treatment (35.7 percent) than in the other treatments. This result is significant at the 5 percent level (p-value $= 0.021$) for LC versus BDM and at the 1 percent level for both LC versus QSR-chances (p-value $= 0.009$) and for LC versus QSR-numbers (p-value $= 0.0001$) based on a two-tailed randomization test. The middle two columns of the table show the directions of deviations when they are not correct but are still on the right side of one-half. For the lottery choice and BDM, the incidence of deviations toward the center and toward the boundary were approximately balanced. In contrast, deviations toward the boundary for the QSR-chances and QSR-numbers occurred about twice as often as deviations toward the center (54.8 percent compared to 25.7 percent and 56.5 percent compared to 26.2 percent). The extremeness that we observed earlier is more than just a tendency to choose boundary rows in the QSR payoff table that has

[33] For example, as can be seen from the sample table in the instructions for the QSR treatments (in the Appendix), the payoffs from reporting 0 chances in 100 of the Red Cup are \$2.00 if the Blue Cup is used and \$0.00 if the Red Cup is used. However, due to rounding with the smaller payoff amounts, reporting 5 chances in 100 of the Red Cup still gives \$2.00 if the Blue Cup is used but \$0.20 if the Red Cup is used. Thus, if there is any uncertainty about the cup being used, it would be rational for an individual to prefer to report 5 chances in 100 rather than 0 chances in 100 (regardless of risk attitude). This feature would presumably lead subjects away from the boundaries as the same flatness appears near 100 chances in 100 of the Red Cup. Despite this, we see a significantly larger number of subjects report boundary probabilities with QSR than with the other elicitation mechanisms. The issue of rounding off payoffs is, of course, a design question that involves tradeoffs. Presenting a table with a long series of digits (representing fractions of a penny) may obscure the overall payoff structure for subjects who are looking at the big picture, although the extra digits provide more accuracy in the flat areas of the quadratic payoff functions.

[34] Decisions were coded as "correct" if they were within $+$ or -1 chance in 100 of the Bayes' rule prediction.

Table 3—Directions of Deviations in Reported Probabilities from Bayes' Predictions when the Bayes' Rule Prediction $\neq$ ½

	"Correct" decisions (+ or −0.01)	Deviations toward center	Deviations toward extremes	Wrong side of 0.5
LC	35.7 percent	31.6 percent	28.0 percent	4.7 percent
BDM	20.0 percent	38.7 percent	33.4 percent	7.9 percent
QSR-chances	16.8 percent*	25.7 percent	54.8 percent	2.7 percent
QSR-numbers	11.1 percent*	26.2 percent	56.5 percent	6.3 percent

*The correct decisions with the QSR will generally differ from Bayes' predictions.

probability labels. The incidence of deviations toward the boundary was significantly more common with both QSR treatments than with either LC (p-value $<= 0.0002$) or BDM (p-value $<= 0.004$) at the 1 percent level using a two-tailed randomization test. The last interesting feature of the breakdown in Table 3 is that the BDM treatment and the QSR-numbers treatment have a larger proportion of decisions on the "wrong side" of one-half, meaning that there were more rounds where subjects even incorrectly predicted which cup was more likely. The QSR-chances treatment has significantly fewer reports on the wrong side of one-half than both BDM at the 5 percent level (p-value $= 0.0317$) and QSR-numbers at the 15 percent level (p-value $= 0.1204$) based on a two-tailed randomization test. This may be due to a lack of understanding with BDM and the absence of probability labels with QSR-numbers.

We are not sure why the quadratic scoring rules tended to yield more extreme beliefs in our experiment, but there are a couple of possibilities that could be investigated in subsequent work. The first possibility involves confusion. As noted above, we see more extreme responses in the QSR-numbers treatment without probability labels, in which some subjects may be confused by seeing a table with an array of numbers that do not have intuitive interpretations. The quadratic scoring rule used by Hollard, Massoni, and Vergnaud (2010) was also implemented with unlabeled decision rows, and about two-thirds of the responses involved extreme (0/100) or uninformed (50/50) choices. Confusion, however, appears to be a partial explanation as it may explain a higher frequency of extreme responses for the QSR-numbers treatment that lacks intuitive probability labels but does not necessarily explain the heightened extremeness in the more easily interpreted QSR-chances treatment with probability labels. Another possibility is the amount of information a subject is asked to process. With BDM, subjects are just asked for a single number and at either stage of the LC, subjects are only required to process ten rows of choices between lotteries. However, for either QSR, subjects are presented with a table that has 100 rows, which may make it more difficult or increase the effort needed by subjects to process and make an accurate decision. A third possibility that involves risk aversion can potentially explain why we do not observe a strong pull-to-center with the QSR in our experiment (although it cannot explain the pull-to-extremes). The subjects in our experiment were paid for each forecast, and a sequence of tasks with small payoffs would tend to reduce the effects of risk aversion and thus reduce the theoretical pull-to-center bias with QSR. In other experiments, when only one forecast is selected ex post at random for payment (as in Armantier and Treich 2013),

that requires that the payoffs be large enough to provide appropriate compensation for the session which can exacerbate the theoretical effects of risk aversion with the QSR. Moreover, the "pull to center" bias Armantier and Treich (2013) observed with QSR was even stronger when payoffs were increased by a factor of 10.

One additional finding relates to the time it takes an individual to make a decision. We electronically recorded the number of seconds between when the subject first saw the submit page and when the subject proceeded to the results page. This decision time was recorded for each round for every individual in the QSR treatments and for the second session of each of the other two treatments.[35] We find that the average decision time in a round across individuals in the LC treatment is 31 seconds which is significantly higher at the 1 percent level (p-value < 0.0001) using a two-tailed randomization test than the QSR-numbers treatment (20 seconds), the QSR-chances treatment (17 seconds), and the BDM treatment (15 seconds). While brevity and speed would certainly be valued in a belief elicitation method since it is often nested within a broader experiment, we feel that 31 seconds to make a decision is not overly restrictive.[36] The longer decision time could be an artifact of procedures (whereby subjects have to click on a radio button in each of the ten rows of a table and then select a more refined choice), or it could be that subjects are taking more time to think through their decisions with the LC menu.[37]

IV. Conclusion

Theory typically places strong restrictions on equilibrium beliefs, and yet it is difficult to infer beliefs from complex decision patterns in experiments. There is a growing literature on direct elicitation of risk preferences, but direct elicitation of beliefs has lagged behind, despite the central role of beliefs in economic theory.[38] The most common approach, the quadratic scoring rule, can be theoretically biased by risk aversion in a manner that has been manipulated by scaling up laboratory payoffs (Armantier and Treich 2013). This bias is due to the penalty for stating a high probability for the event not observed, a risk that can be eliminated by equalizing reported probabilities. Moving in the opposite direction, toward low or minimal incentives, has resulted in a higher incidence of nonsensical or inconsistent responses.[39]

[35] The decision time was not recorded for the first session of the LC and BDM treatments simply because we had not thought of that perspective and thus the programming to record the time was incomplete. If we restrict analysis to the second session of each treatment, the average decision time for QSR would change only slightly from 17 to 19 seconds.

[36] However, it would be interesting to determine if there was improved accuracy if subjects were given BDM or QSR choice problems twice compared to only one choice in the LC method.

[37] We did not record how much of the decision time was used up by re-choosing, or how often this occurs. Such re-choosing could explain the longer decision times for LC, but it is worth noting that multiple crossover attempts were quite infrequent in hand-run versions of the LC treatment done for the Tulman (2013) dissertation on voting that is discussed in the conclusion. The subjects who did provide multiple crossovers (less than 5 percent) were forced to re-choose.

[38] See Schotter and Trevino (2014) for an up-to-date survey of the literature on belief elicitation.

[39] For example, Grether (1992) reports about three times as many "nonsense or incoherent responses" without monetary incentives. Similarly, Armantier and Treich (2013) observe more extreme responses (near 0 or 1) and responses on the wrong side of one-half with hypothetical payoffs.

One solution to risk aversion or risk-preference biases is to transform scoring rule payoffs into lottery tickets for a fixed money prize. This "binary lottery" procedure has reduced prediction errors in Bayesian information processing tasks, but has shown mixed results in its ability to reliably induce risk neutrality in other contexts (Prasnikar 1993; Selten, Sadrieh, and Abbink 1999). An alternative is to use Becker-DeGroot-Marschak methods based on the intuition behind second price auctions. The BDM method, however, is notorious for being confusing to subjects.

The lottery choice menu used in this paper is structured to be more intuitive than BDM but retain invariance to risk attitudes. The two-stage menu lets subjects choose between an event-contingent payoff and a payoff determined by an exogenous probability that changes from row to row, where the "crossover" in the second stage determines the elicited belief. Since the binary payoffs for the event lottery and the random lottery are synchronized in the choice menu, this method theoretically induces truthful elicitation of beliefs regardless of risk attitudes, a feature that distinguishes it from QSR and approaches that have more than two possible payoffs, and hence that require an extra layer of risk preference measurement and adjustment. Such de-biasing procedures are not commonly used because they may introduce measurement error in one dimension in an attempt to correct measurement error in another.

This paper presents an experimental test of the synchronized two-stage choice menu for determining a subject's beliefs about the "chances in 100" that an event will be observed. The experiment is done in the context of a Bayesian information processing task with a symmetric prior and some trials with no sample information, features that are intended to create a less stressful belief formation environment. We also test the performance of a BDM procedure and the standard QSR, with and without probability labels. All procedures perform well in the sense of producing median beliefs that roughly track the true Bayes' probabilities. The most obvious deviation from Bayes' predictions is that the QSR treatments exhibit a pull toward the more extreme probabilities when the event probabilities differ from one-half. The proportion of correct Bayes' responses (within plus or minus 0.01 of the true probability) is significantly higher with the lottery choice menu than the other methods. The lottery choice menu has significantly lower average forecast errors than the QSR without probability labels, and it exhibits marginally lower average absolute forecast errors than BDM, both for all trials and for trials with no information, where what is being elicited is just the prior based on the simulated die throw.

Subjects take about 30 seconds longer to produce a complete set of responses with the lottery choice menu, but we believe that the process of making row-by-row binary comparisons promotes careful thinking. Furthermore, the ordered structure of row choices, with a second page for fine tuning, allows subjects to focus on the big picture and then make final adjustments. In conclusion, the synchronized lottery choice menu is relatively easy to comprehend, and it performs as well or better than the more complex BDM procedure. The lottery choice method performs much better than the QSR when probability labels are not used, and it does not suffer from any theoretical risk aversion bias that makes it awkward or inappropriate to use intuitive probability labels for the QSR payoff table.

In our view, the two-stage lottery choice method has some theoretical and/or empirical advantages over other commonly used elicitation techniques, including BDM and QSR. However, there are other mechanisms that have not been directly compared to the lottery choice method and these comparisons serve as a direction for future research. For example, Hao and Houser (2010) use an English clock auction based implementation of BDM that outperforms the standard BDM procedure. In addition, the lottery choice method has not been compared directly with a QSR method combined with a de-biasing procedure or converted to a binary lottery or with an un-incentivized procedure.

Although comparisons of elicitation procedures under controlled conditions are suggestive, the real test is in terms of usefulness in laboratory and field studies of economic behavior. Synchronized lottery choice menus are easy to explain, and the results can be used in field settings without additional econometric analysis. These two-stage menus were successfully implemented with paper decision sheets in two recent doctoral dissertations. In a laboratory study of voting games with randomly voting costs, Tulman (2013) found that voter turnout decisions were correlated with elicited beliefs about being pivotal. In a threshold voluntary contributions experiment, Schreck (2013) found that contributions were strongly correlated with elicited beliefs about whether the person's contribution would trigger a threshold match. This dissertation provides a clean set of instructions and menus for hand-run elicitation (Sehreck 2013, 82–86). If we were solely interested in how beliefs changed across treatments, a systematic bias would be less concerning because any bias would be present across all treatments using the same elicitation procedure. However, as was the case in both of these applications, beliefs are key components of economic models and it is thus important to elicit accurate (unbiased) beliefs in order to truly test the link between beliefs and decisions even within a given treatment. A practical way of eliciting beliefs without the need to correct for risk preferences should be useful in a wide range of laboratory and field studies.

In both of the dissertations discussed above, beliefs were elicited *after* the final round of decisions (but before results were announced). This procedure serves to avoid any game decision-making bias resulting from the process of elicitation itself, but does place a limit on the amount and type of data that can be collected.[40] It should be noted, however, that employing a belief elicitation mechanism in the middle of an experiment can potentially alter game decisions. For example, the mere act of eliciting beliefs about other subjects' behavior may cause individuals to focus on those beliefs more strongly and act more in accordance with theory based on those probabilities than they otherwise would have. Previous research on the effect of belief elicitation conducted either before or simultaneously with game decisions is mixed. Some studies (i.e., Croson 2000; Hoffman 2013; Zelmer 2003) find that elicitation of beliefs causes subject behavior to more closely mimic theoretical predictions, while others find a significant effect of belief elicitation on

[40] Elicitation of beliefs after game decisions have been made can still potentially affect the elicited belief itself. Hoffman (2013) finds that elicited beliefs are similar whether they are elicited simultaneously with decisions or after all decisions made while, of course, actual game decisions can only be affected with belief elicitation in the middle of an experiment.

behavior but that subject behavior is pushed away from theoretical predictions (i.e., Gächter and Renner 2010). Still other studies find little or no effect of elicitation on decision-making (i.e., Wilcox and Feltovich 2000; Nyarko and Schotter 2002; Costa-Gomes and Weizsäcker 2008). This potential bias is, in general, a concern with any belief elicitation mechanism. However, it is possible that the two-stage menu procedure exacerbates this issue due to the extra time involved and repeated pairings of lottery choices. It will be important for future research to continue to investigate the presence and degree of impact of this bias both across elicitation techniques and across types of games.

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